Jiayi (Chesley) Guo

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Education

University of California, Berkeley **2025 – Dec 2026 (Expected)**
Master of Development Engineering (MDE), AI/Data Analytics Track; GPA: 3.82 *Berkeley, CA*

- **Core Courses:** Experiments & Causal Inference; Research Design & Applications for Data and Analysis; Big Data & Development; Digital Transformation of Development; Design, Evaluate & Scale Development Technologies; Critical Systems of Development; Front-End Web Architecture; Public Interest Technology Clinic
- **Honors:** NSF Research Traineeship (NRT) in Digital Transformation of Development (DToD), 2025

University of Illinois Urbana-Champaign **2023 – 2024**
Bachelor of Science (BS) in Information Sciences, Data Science Track; GPA: 3.91 *Urbana, IL*

- **Core Courses:** Data Visualization; Research Design in Information Sciences; Programming for Information Problems; Analytical Foundations of Information Problems; Reading and Writing Data; Introduction to HCI; Ethics & Policy for Data Science; Data Structures (transfer credit); Advanced Software Design (transfer credit)
- **Honors:** Edmund J. James Scholar (2023–2024)
  Dean's List (Fall 2023)
  Phi Kappa Phi Honor Society (2024)
  WeCU Scholar, 200+ hours of volunteer service (2024)
  Top 3 Finalist, Equitable Access to Healthcare Challenge, MIT Policy Hackathon (2023)
  Competition Expert, Kaggle (2023)

Xi'an Jiaotong-Liverpool University **2020 – 2022**
Undergraduate study in Bioinformatics (two years); transferred to UIUC *Suzhou, China*

- **Core Courses:** Cellular & Molecular Biology; Introduction to Biochemistry; Introduction to Evolution & Genetics; Algorithmic Foundations & Problem Solving; Introduction to Databases; Introduction to Programming in Java; Experimental Design & Analysis; Introduction to Statistics; Linear Algebra; Multivariable Calculus
- **Honors:** Meritorious Winner, Interdisciplinary Contest in Modeling (ICM, 2022)
  National Second Prize, China Undergraduate Mathematical Contest in Modeling (2021)
  District Third Prize, Asia-Pacific Mathematical Contest in Modeling (2020)

Publications

- **Guo, J.** Retracted but not forgotten: A matched-control audit of how consumer large language models handle retracted clinical trials. *Submitted to Scientometrics.*
- **Guo, J.**, & Hughes, A. The effect of automated speed enforcement on driving behavior: Evidence from Oakland's speed safety camera pilot. *Working paper.*
- **Guo, J.**, & Agogino, A. M. Detected once, counted twice? Selection-aware simulation-based inference for landfill methane from sparse airborne surveys. *Working paper.*
- Lu, H., Gu, L., **Guo, J.**, Xu, L., et al. (2023). Machine learning classification and recovery prediction of COVID-19 cough sounds. *SSRN (Preprints with The Lancet).* SSRN 4439141
- Shao, Y., **Guo, J.**, Li, Z., Yan, W., & Zhu, Z. (2022). Optimization of seaport companies' D&A systems based on system dynamics and evaluation models. *Proceedings of SPIE*, 12340, 1234003. doi:10.1117/12.2652356

Additional Training

MITx MicroMasters Program in Statistics and Data Science **2024**
Completed all four courses with verified certificates: Probability; Statistics; ML with Python; Data Analysis *Online*

Oxford Machine Learning Summer School (OxML) **2023**
ML × Health Track *Oxford, UK*

SJTU–THU–PKU Computer Science International Summer School **2023**
Courses: Artificial Intelligence; Cryptography in Blockchain *Shanghai, China*

Healthcare Big Data Summer Program, Duke Kunshan University **2022**
Courses: Health Data Analytics; ML in Health Sciences; Causal Inference & RCTs; Real-World Evidence *Kunshan, China*

Data Analysis for Life Sciences Professional Certificate, Harvard University **2022**
Courses: Statistics and R; Linear Models; Statistical Inference; High-Dimensional Data Analysis *Online*

Independent Research

Retracted Clinical Trials and Consumer LLMs **Aug 2026 – Present**
Sole author; manuscript submitted to Scientometrics *Berkeley, CA*

- Designed a matched-control audit of whether consumer LLMs present retracted clinical trials as valid evidence, pairing 100 retracted trials (Retraction Watch) with non-retracted controls matched on topic, year and design.
- Collected 3,600 responses from the free-tier default models of ChatGPT, Gemini and Claude across 3 question framings and 3 conditions (baseline, retraction notice, web search); coded them with an LLM coder validated against hand coding ($\kappa = 0.96$) and analyzed them with GEE models clustered by trial pair.
- Found that ∼2/3 of answers presented retracted findings as valid, with no difference from matched controls (OR 0.89, n.s.); a retraction notice cut errors to ≤2%, while web search had mixed effects (GPT 66%→35%, Gemini 66%→82%). Code and data: github.com/chesleyguo/llm-retracted-clinical-evidence

Research Experience

School of Information, University of California, Berkeley **Aug 2026 – Present**
Student Team Data Scientist; Supervisor: Prof. Alex Hughes *Berkeley, CA*

- Serving on an automated speed enforcement causal impact study in collaboration with the Oakland Department of Transportation (OakDOT) and its Data Science team, tracking anonymized vehicle violation histories over time to analyze repeat-offense dynamics and long-term driver compliance patterns.
- Building Difference-in-Differences models on multi-year panel data to isolate the net causal impact of speed camera deployments on traffic speed distributions and severe speeding incidents.
- Processing and merging 3 years of high-frequency speed count data and automated enforcement violation records within a secure GCP environment under CPRA/CCPA compliance.

College of Engineering, University of California, Berkeley **Jul 2026 – Present**
Research Assistant; Supervisor: Prof. Alice Agogino *Berkeley, CA*

- Developing a selection-aware simulation-based inference (SBI) framework that estimates facility-level, time-averaged landfill methane point-source emissions, with calibrated uncertainty, from sparse airborne surveys (AVIRIS-NG, GAO; 107 U.S. landfills, 593 flight days).
- Built a reproducible pipeline that reconciles airborne plume observations with EPA GHGRP Subpart HH reports and LMOP facility records, and quantifies the probability that a facility's observed point-source emissions exceed its reported methane.
- Showed in simulation that averaging only detected overpasses can overstate a facility's time-averaged emissions by more than 2×; the framework corrects this by modeling non-detections, instrument-specific detection limits, and survey selection.
- Previously audited EPA's Landfill Methane Outreach Program register (2,641 landfills) with leave-state-out validation and released an interactive public explorer.
- Built a landfill methane dashboard now used by the product team at Squishy Robotics, the lab's startup partner.

OpenTrust Lab, Shanghai Artificial Intelligence Laboratory **Jun 2024 – Aug 2024**
Research Intern; Supervisor: Dr. Jing Shao, Research Scientist *Shanghai, China*

- Assisted with safety evaluations of large language models on the lab's benchmarks (SALAD-Bench, MM-SafetyBench).
- Supported organization of the WAIC 2024 Security Forum and the TiFA Workshop at ICML 2024.

School of Information Sciences, University of Illinois Urbana-Champaign **Jan 2024 – May 2024**
Research Assistant; Supervisor: Prof. Yun Huang *Urbana, IL*

- Conducted literature review and comparative analysis on cross-cultural Human-Computer Interaction (HCI) designs for academic information-seeking behaviors.
- Assisted in the conceptual design and requirement specification for an AI-powered chatbot tailored to enhance accessibility and efficiency in academic search and literature discovery for a global user base.

- Proposed and outlined key features for the chatbot, including multilingual support, context-aware query interpretation, and integration with digital library APIs, to address diverse user needs in scholarly environments.

Vanke School of Public Health, Tsinghua University **Nov 2022 – Mar 2023**
Research Assistant; Supervisor: Prof. Kun Tang *Beijing, China (Remote)*

- Collected, cleaned, and processed textual data on medical cooperation memorandums and IHME disease burden metrics (DALYs) for Belt and Road countries since 2016 (Africa, South America, Southeast Asia) to support the team's downstream analysis.

Global Health Institute, Duke Kunshan University **Mar 2022 – Aug 2022**
Health Research Data Analyst Intern; Supervisor: Prof. Chenkai Wu *Kunshan, China*

- Conducted a comprehensive analysis using descriptive statistics, principal component analysis, and multinomial regression to examine the relationship between 249 metabolites from the UK Biobank and the impact of sarcopenia on aging; built mortality risk predictions using the Cox proportional hazards model.

Shanghai Key Laboratory of Multidimensional Information Processing **Oct 2021 – Dec 2022**
Research Assistant; Supervisor: Prof. Jiangang Chen, East China Normal University *Shanghai, China*

- Collected cough acoustics data from 78 subjects at a tertiary hospital in Shanghai; performed data processing and cleaning to support the identification of COVID-19 patients.
- Leveraged voice activity detection, noise reduction, and data augmentation techniques to improve data quality, expanding the dataset to 535 audio samples; extracted diverse audio features and their statistical characteristics using `librosa` and `scipy`.
- Constructed prognostic models integrating multiple machine learning algorithms (e.g., Random Forest, XGBoost, SVM) to balance detection rate and prediction of time to recovery; reduced prognostic time error to within 3 days and achieved 93.3% accuracy with an ensemble XGBoost and Random Forest model.

Work Experience

Tencent **May 2025 – Aug 2025**
Product Manager Intern *Shanghai, China*

- Analyzed product usage data for Tencent's Real-Time Communication (RTC) suite (e.g., Conversational AI, Call, Conference, Live Streaming) to identify and prioritize feature improvements for a global user base.
- Ran A/B tests across marketing campaigns to compare the effectiveness of content strategies.

ByteDance (TikTok) **Nov 2024 – Jan 2025**
AI Strategy Product Manager Intern *Shanghai, China*

- Worked with algorithm engineers to improve video understanding and text generation LLMs, focusing on description accuracy and hallucination reduction.
- Drafted product requirement documents (PRDs) and defined core functionalities for LLMs, while designing and refining evaluation metrics such as accuracy, semantic relevance, and logical reasoning.

CSC Financial Co., Ltd. **Jul 2023 – Aug 2023**
Financial Planning and Analysis Analyst Intern *Changzhou, China*

- Built financial models and forecasts to support budget creation and the firm's strategic planning process.
- Conducted competitor and market analyses to identify growth opportunities, and presented findings and recommendations to senior management.

Teaching & Mentoring

- **Research Mentor**, Undergraduate Research Apprentice Program (URAP), UC Berkeley 2026 – Present
  Mentoring 2 undergraduates on their independent research projects in AI/ML for methane monitoring.
- **Senior Mentor**, Xi'an Jiaotong-Liverpool University Affiliated School 2022 – 2023
  Taught STEM enrichment courses to a class of ∼50 high school students.
- **Student Representative**, School of Science, Xi'an Jiaotong-Liverpool University 2021 – 2022
  Mentored junior undergraduates on coursework and academic planning.